IMAGES OF UNIVERSALLY CLOSED AND PROPER MORPHISMS INTO A SEPARATED SCHEME

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-morphisms/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot. Nothing was added, removed or redefined, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

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Allowed standard prerequisites (editorial).

- The diagonal-like morphism into the fibre product is a closed immersion when the target is separated. Schemes, Lemma \ref{schemes-lemma-semi-diagonal}, cited at proof lines 9968-9970. Makes the first factor a closed immersion.
- Closed immersions are proper. Lemma \ref{lemma-closed-immersion-proper}, cited at proof line 9971. Gives properness of the first factor.
- The second factor is a base change of the given morphism. used at proof lines 9972-9979. Carries universal closedness and properness across.

Source reference. Original lemma 01W6: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2471–2471. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 01KS: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2061–2061. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

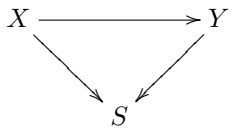
Source reference. Original lemma 01W5: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2470–2470. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Date: stacks-project:morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 01W4: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2469–2469. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 01W3: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2468–2468. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Suppose given a commutative diagram of schemes*



with Y separated over S.

(1) *If $X \to S$ is universally closed, then the morphism $X \to Y$ is universally closed.*
(2) *If X is proper over S, then the morphism $X \to Y$ is proper.*

In particular, in both cases the image of X in Y is closed.

Proof. Assume that $X \to S$ is universally closed (resp. proper). We factor the morphism as $X \to X \times_S Y \to Y$. The first morphism is a closed immersion, see Schemes, Lemma 01KS. Hence the first morphism is proper (Lemma 01W5). The projection $X \times_S Y \to Y$ is the base change of a universally closed (resp. proper) morphism and hence universally closed (resp. proper), see Lemma 01W4. Thus $X \to Y$ is universally closed (resp. proper) as the composition of universally closed (resp. proper) morphisms (Lemma 01W3). □